

DEPARTMENT OF MATHEMATICS
UNIVERSITY OF MARYLAND
GRADUATE WRITTEN EXAM

January 2018

ALGEBRA (Ph.D. Version)

Instructions to the student

- Answer all six questions; each will be assigned a grade from 0 to 10.
 - Use a different booklet for each question. Write the problem number and your **code number** (**not** your name) on the outside of the booklet.
 - Keep scratch work on separate pages in the same booklet.
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- Let G be a group of order $168 = 3 \cdot 8 \cdot 7$, and assume that G has no nontrivial normal subgroups.
 - Show that G has eight Sylow 7-subgroups.
 - Let H be a Sylow 7-subgroup of G and let $N_G(H) = \{g \in G \mid gHg^{-1} = H\}$. Show that $|N_G(H)| = 21$.
 - Show that G has no subgroup of order 14. (*Hint*: There is no element of order 2 in $N_G(H)$.)
- Let A be an $n \times n$ matrix with entries in the complex numbers $\mathbb{C}$. Show that the following two statements are equivalent:

i. A is diagonalizable

ii. For each $x \in \mathbb{C}$,

$$\text{Image}(xI - A) + \text{Ker}(xI - A) = \mathbb{C}^n,$$

where I is the $n \times n$ identity matrix.

- Let R be a commutative ring with 1 and let Q be a proper ideal of R . We say that Q is *primary* if it satisfies the following condition: Let $x, y \in R$ with $xy \in Q$. If $x \notin Q$ then $y^n \in Q$ for some positive integer n (depending on y). Define the radical of Q by

$$\text{rad}(Q) = \{x \in R \mid x^n \in Q \text{ for some positive integer } n\}.$$

Note that $\text{rad}(Q)$ is also a proper ideal of R (do not prove this).

- Prove that if Q is primary then $\text{rad}(Q)$ is a prime ideal.
 - Suppose R is a principal ideal domain. Find all primary ideals of R (prove that all primary ideals are on your list, and prove that all ideals on your list are primary).
- Let R be a commutative ring with 1 and let M be a nonzero Noetherian R -module. Let
$$I = \{r \in R \mid rM = 0\}.$$
 - Show that there is an injection $R/I \rightarrow M^n$ for some $n \geq 1$.
 - Show that R/I is a Noetherian ring.
 - Let L/K be a finite Galois extension of fields and let $\alpha \in L$. Suppose there is an automorphism $\sigma \in \text{Gal}(L/K)$ such that $\sigma(\alpha) = \alpha + 1$.
 - Show that K has characteristic p for some prime $p > 0$.
 - If $L = K(\alpha)$, show that σ has order p .

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- (c) Suppose σ generates $\text{Gal}(L/K)$. Show that $\alpha^p - \alpha \in K$.
6. Let G be a finite group and let $\rho : G \rightarrow \text{GL}_n(\mathbb{C})$ be a representation of G . Let χ be the character of ρ .
- (a) Define the matrix $M = \sum_{g \in G} \rho(g)$. Show that if $M \neq 0$ then there is a nonzero vector v such that $\rho(g)v = v$ for all $g \in G$.
- (b) Suppose $\sum_{g \in G} \chi(g) = 0$. Show that $M = 0$.